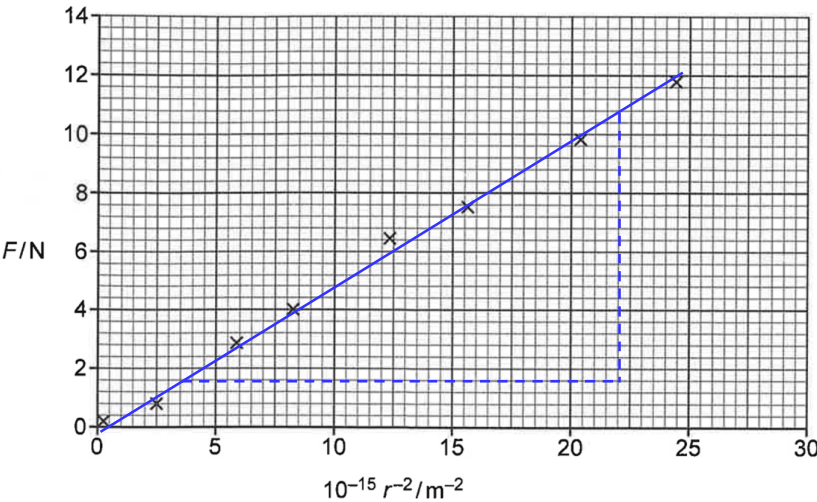


2(a)(i)	Geostationary orbit
2(a)(ii)	The satellite is moving in a circular orbit. The direction of motion of the satellite is constantly changing. Therefore the velocity of the satellite is constantly changing. Therefore the satellite is accelerating.
2(b)(i)	$T = pr^q$ $\lg T = \lg p + q \lg r$ Plot a graph of $\lg T$ against $\lg r$. The gradient of the graph is equal to q. The y-intercept of the graph is equal to $\lg p$. Therefore $p = 10^{\text{y-intercept}}$
2(b)(ii)	For Uranus: $\lg 84 = \lg 1.0 + 1.5 \lg r$ For Earth: $\lg 1 = \lg 1.0 + 1.5 \lg \text{AU}$ Therefore $r = 19.18 \text{ AU} \approx 19 \text{ AU}$

2(c)(i)	 Fig. 2.1
2(c)(ii)	Using $(3.50 \times 10^{-15}, 1.6)$ and $(22.00 \times 10^{-15}, 10.8)$, $\text{gradient of graph} = \frac{10.8 - 1.6}{(22.00 - 3.50) \times 10^{-15}} = 4.973 \times 10^{14} \approx 5.0 \times 10^{14}$ Unit of gradient = (Unit of F)(Unit of r^2) = N m^2
2(c)(iii)	Density $\rho = \frac{m}{l^3} = \frac{1200}{(10.0)^3} = 1.20 \text{ g cm}^{-3}$ Percentage uncertainty $\frac{\Delta \rho}{\rho} \times 100\% = \left[\frac{\Delta m}{m} + 3 \frac{\Delta l}{l} \right] \times 100\% = \left[\frac{10}{1200} + 3 \left(\frac{0.1}{10.0} \right) \right] \times 100\% = 3.83\% \approx 4\%$
3(a)	The moment of a force about a point is the product of the force and the